

Strong and Weak Forms of Solid Mechanics: Spatial, Referential, and Total Lagrangian Formulations

1 Kinematics and Notation

Let $\Omega_R \subset \mathbb{R}^d$ denote the reference (material) configuration of a body (referential quantities carry the subscript R) with boundary $\partial\Omega_R = \Gamma_R^D \cup \Gamma_R^N$, $\Gamma_R^D \cap \Gamma_R^N = \emptyset$, and let Ω_t denote the current (spatial) configuration at time t with boundary $\partial\Omega_t = \Gamma_t^D \cup \Gamma_t^N$. The motion

$$\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t), \quad \mathbf{X} \in \Omega_R, \quad (1)$$

maps material points $\mathbf{X}$ to spatial points $\mathbf{x}$. The displacement, deformation gradient, and Jacobian are

$$\mathbf{u}(\mathbf{X}, t) = \boldsymbol{\varphi}(\mathbf{X}, t) - \mathbf{X}, \quad \mathbf{F} = \text{Grad } \boldsymbol{\varphi} = \mathbf{I} + \text{Grad } \mathbf{u}, \quad J = \det \mathbf{F} > 0, \quad (2)$$

where $\text{Grad}(\cdot) = \partial(\cdot)/\partial\mathbf{X}$ and $\text{grad}(\cdot) = \partial(\cdot)/\partial\mathbf{x}$ denote referential and spatial gradients, respectively. The right Cauchy–Green tensor and the Green–Lagrange strain are

$$\mathbf{C} = \mathbf{F}^\top \mathbf{F}, \quad \mathbf{E} = \frac{1}{2} (\mathbf{C} - \mathbf{I}) = \frac{1}{2} \left(\text{Grad } \mathbf{u} + \text{Grad}^\top \mathbf{u} + \text{Grad}^\top \mathbf{u} \text{ Grad } \mathbf{u} \right). \quad (3)$$

The velocity is $\mathbf{v} = \dot{\boldsymbol{\varphi}} = \dot{\mathbf{u}}$ (material time derivative). Mass conservation relates the referential and spatial densities by $\rho_R = J\rho$.

The stress measures used below are the Cauchy stress $\boldsymbol{\sigma}$ (force per unit *current* area), the first Piola–Kirchhoff stress $\mathbf{P}$ (force per unit *reference* area), and the second Piola–Kirchhoff stress $\mathbf{S}$, related through the Piola transform

$$\mathbf{P} = J\boldsymbol{\sigma}\mathbf{F}^{-\top}, \quad \mathbf{S} = \mathbf{F}^{-1}\mathbf{P} = J\mathbf{F}^{-1}\boldsymbol{\sigma}\mathbf{F}^{-\top}, \quad \boldsymbol{\sigma} = \frac{1}{J}\mathbf{P}\mathbf{F}^\top = \frac{1}{J}\mathbf{F}\mathbf{S}\mathbf{F}^\top. \quad (4)$$

Note that $\mathbf{S} = \mathbf{S}^\top$ and $\boldsymbol{\sigma} = \boldsymbol{\sigma}^\top$ (balance of angular momentum), while $\mathbf{P}$ is in general unsymmetric, satisfying $\mathbf{P}\mathbf{F}^\top = \mathbf{F}\mathbf{P}^\top$.

2 Strong Form

2.1 Spatial (Eulerian / updated) form

Balance of linear momentum on the current configuration Ω_t reads: find $\mathbf{u}$ (equivalently the motion $\boldsymbol{\varphi}$) such that

$$\rho \dot{\mathbf{v}} = \text{div } \boldsymbol{\sigma} + \rho \mathbf{b} \quad \text{in } \Omega_t, \quad (5a)$$

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \Gamma_t^D, \quad (5b)$$

$$\boldsymbol{\sigma} \mathbf{n} = \bar{\mathbf{t}} \quad \text{on } \Gamma_t^N, \quad (5c)$$

$$\mathbf{u}(\cdot, 0) = \mathbf{u}_0, \quad \mathbf{v}(\cdot, 0) = \mathbf{v}_0 \quad \text{in } \Omega_R, \quad (5d)$$

where ρ is the current mass density, $\mathbf{b}$ the body force per unit mass, $\mathbf{n}$ the outward unit normal on $\partial\Omega_t$, $\bar{\mathbf{t}}$ the prescribed traction per unit current area, and $\dot{\mathbf{v}} = \partial\mathbf{v}/\partial t + (\text{grad } \mathbf{v})\mathbf{v}$ the material acceleration. For quasi-statics the inertial term $\rho \dot{\mathbf{v}}$ is dropped. Angular momentum balance requires $\boldsymbol{\sigma} = \boldsymbol{\sigma}^\top$.

2.2 Referential (Lagrangian) form

Pulling (5) back to the reference configuration Ω_R using the Piola transform (4) and Nanson's formula $\mathbf{n} da = J\mathbf{F}^{-\top}\mathbf{N} dA$, the momentum balance becomes: find $\mathbf{u} : \Omega_R \times [0, T] \rightarrow \mathbb{R}^d$ such that

$$\rho_R \ddot{\mathbf{u}} = \text{Div } \mathbf{P} + \rho_R \mathbf{b} \quad \text{in } \Omega_R, \quad (6a)$$

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \Gamma_R^D, \quad (6b)$$

$$\mathbf{P} \mathbf{N} = \bar{\mathbf{T}} \quad \text{on } \Gamma_R^N, \quad (6c)$$

$$\mathbf{u}(\cdot, 0) = \mathbf{u}_0, \quad \dot{\mathbf{u}}(\cdot, 0) = \mathbf{v}_0 \quad \text{in } \Omega_R, \quad (6d)$$

where $\rho_R = J\rho$ is the reference density, $\mathbf{N}$ the outward unit normal on $\partial\Omega_R$, and $\bar{\mathbf{T}} = \bar{\mathbf{t}}(da/dA)$ the prescribed nominal traction (force per unit *reference* area). All derivatives are taken with respect to $\mathbf{X}$ over the fixed domain Ω_R ; this is the natural setting for a total Lagrangian discretization.

The system is closed by a constitutive relation. For a hyperelastic material with stored energy density $\Psi(\mathbf{F})$ (per unit reference volume),

$$\mathbf{P} = \frac{\partial \Psi}{\partial \mathbf{F}}, \quad \text{or equivalently} \quad \mathbf{S} = 2 \frac{\partial \Psi}{\partial \mathbf{C}} = \frac{\partial \Psi}{\partial \mathbf{E}}. \quad (7)$$

3 Weak Form: Total Lagrangian Formulation

Define the trial and test spaces over the *fixed* reference domain,

$$\mathcal{S}_t = \left\{ \mathbf{u} \in [H^1(\Omega_R)]^d : \mathbf{u} = \bar{\mathbf{u}} \text{ on } \Gamma_R^D \right\}, \quad \mathcal{V} = \left\{ \mathbf{w} \in [H^1(\Omega_R)]^d : \mathbf{w} = \mathbf{0} \text{ on } \Gamma_R^D \right\}. \quad (8)$$

Taking the inner product of (6a) with $\mathbf{w} \in \mathcal{V}$, integrating over Ω_R , and using integration by parts together with $\mathbf{P} \mathbf{N} = \bar{\mathbf{T}}$ on Γ_R^N and $\mathbf{w} = \mathbf{0}$ on Γ_R^D , yields the total Lagrangian weak form:

Find $\mathbf{u}(\cdot, t) \in \mathcal{S}_t$ such that, for all $\mathbf{w} \in \mathcal{V}$,

$$\boxed{\int_{\Omega_R} \rho_R \ddot{\mathbf{u}} \cdot \mathbf{w} dV + \int_{\Omega_R} \mathbf{P}(\mathbf{F}) : \text{Grad } \mathbf{w} dV = \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} dV + \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} dA} \quad (9)$$

with $\mathbf{F} = \mathbf{I} + \text{Grad } \mathbf{u}$. Every integral is evaluated on the reference configuration, which never changes during the computation—the defining feature of the total Lagrangian formulation.

3.1 Equivalent form in the second Piola–Kirchhoff stress

Substituting $\mathbf{P} = \mathbf{F}\mathbf{S}$ and noting that $\mathbf{F}\mathbf{S} : \text{Grad } \mathbf{w} = \mathbf{S} : \mathbf{F}^\top \text{Grad } \mathbf{w} = \mathbf{S} : \delta\mathbf{E}[\mathbf{w}]$ (the last step using the symmetry of $\mathbf{S}$), where

$$\delta\mathbf{E}[\mathbf{w}] = \frac{1}{2} \left(\mathbf{F}^\top \text{Grad } \mathbf{w} + \text{Grad}^\top \mathbf{w} \mathbf{F} \right) \quad (10)$$

is the variation of the Green–Lagrange strain in the direction $\mathbf{w}$, the weak form (9) may equivalently be written

$$\int_{\Omega_R} \rho_R \ddot{\mathbf{u}} \cdot \mathbf{w} \, dV + \int_{\Omega_R} \mathbf{S}(\mathbf{E}) : \delta \mathbf{E}[\mathbf{w}] \, dV = \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} \, dV + \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} \, dA. \quad (11)$$

This is the form commonly used with $(\mathbf{S}, \mathbf{E})$ constitutive pairs (e.g. St. Venant–Kirchhoff or models expressed in terms of $\mathbf{C}$).

3.2 Remarks

- **Quasi-statics.** Dropping the inertial term in (9) gives the virtual work statement $\int_{\Omega_R} \mathbf{P} : \text{Grad } \mathbf{w} \, dV = \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} \, dV + \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} \, dA$, i.e. internal virtual work equals external virtual work.
- **Nonlinearity and linearization.** The weak form is nonlinear in $\mathbf{u}$ through $\mathbf{P}(\mathbf{F})$ (and geometrically through $\mathbf{F}$). A Newton scheme requires the consistent tangent $\mathbb{A} = \partial \mathbf{P} / \partial \mathbf{F} = \partial^2 \Psi / \partial \mathbf{F} \partial \mathbf{F}$, giving the directional derivative $D_{\Delta \mathbf{u}} \int_{\Omega_R} \mathbf{P} : \text{Grad } \mathbf{w} \, dV = \int_{\Omega_R} \text{Grad } \mathbf{w} : \mathbb{A} : \text{Grad } \Delta \mathbf{u} \, dV$; in the $(\mathbf{S}, \mathbf{E})$ form this splits into the familiar material and geometric stiffness contributions.
- **Follower loads.** A traction prescribed per unit *current* area (e.g. pressure $\bar{\mathbf{t}} = -p \mathbf{n}$) becomes configuration-dependent when pulled back: $\bar{\mathbf{T}} = -p J \mathbf{F}^{-\top} \mathbf{N}$, and then contributes to the tangent stiffness. The form above assumes $\bar{\mathbf{T}}$ is a dead (nominal) load.
- **Relation to the spatial weak form.** Pushing (9) forward with $\mathbf{P} : \text{Grad } \mathbf{w} = J \boldsymbol{\sigma} : \text{grad } \mathbf{w}$ and $dv = J \, dV$ recovers the updated/spatial weak form $\int_{\Omega_t} \rho \dot{\mathbf{v}} \cdot \mathbf{w} \, dv + \int_{\Omega_t} \boldsymbol{\sigma} : \text{grad } \mathbf{w} \, dv = \int_{\Omega_t} \rho \mathbf{b} \cdot \mathbf{w} \, dv + \int_{\Gamma_t^N} \bar{\mathbf{t}} \cdot \mathbf{w} \, da$; the two statements are identical, differing only in the configuration on which integrals are evaluated.